Final Project: Physics-Guided Learning for NextG Channel Modeling

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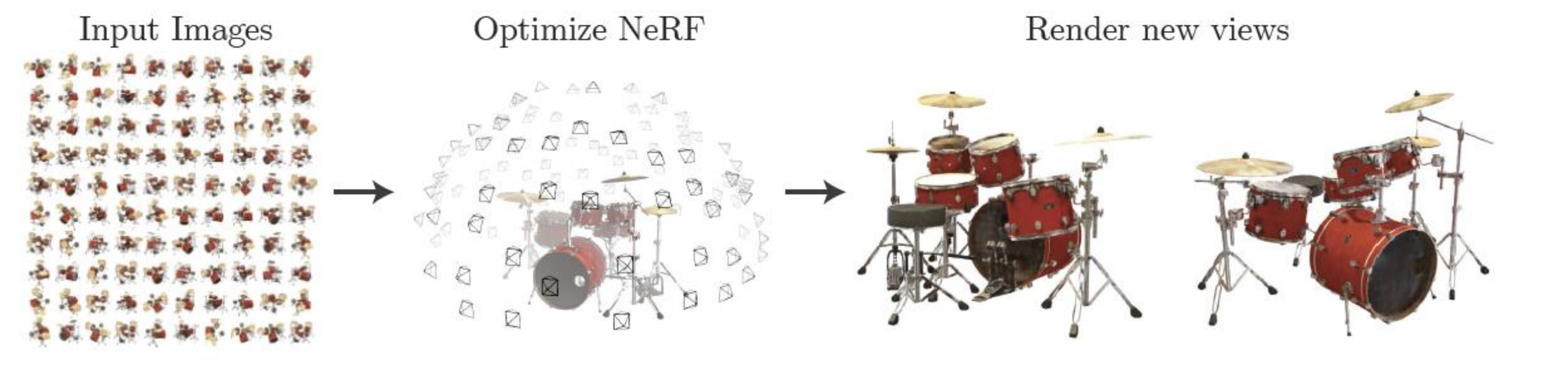
Background

Idea: RF simulation software have their patented, closed-source model that are very accurate to simulate RF channel in complex scenes, we cannot know their equations, and they require input of the exact geometry of the environment to work, yet we can use MLP+inaccurate RF model, to get accurate channel prediction after training, and we do not need scene geometry knowledge or only need partial knowledge.

Nvidia's SIONNA raytracing RF channel simulation software

Background

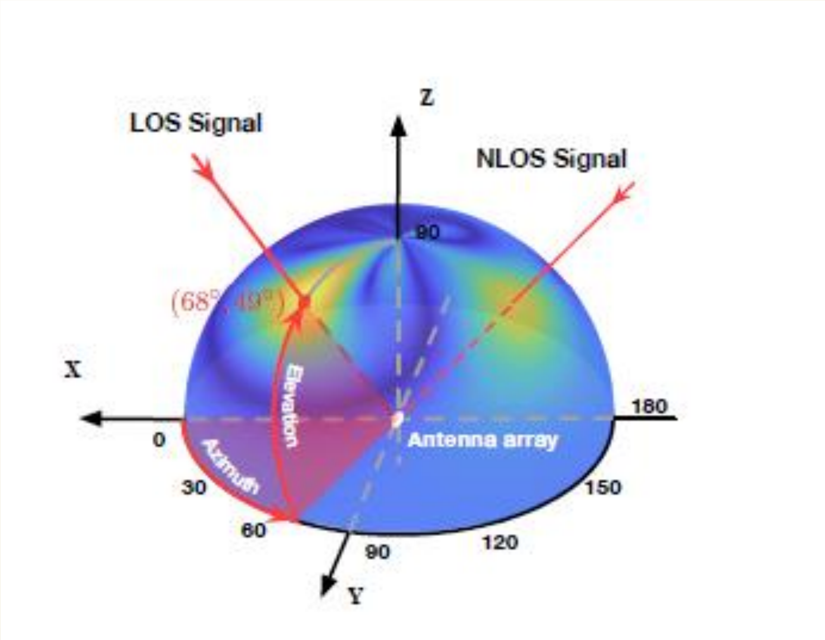


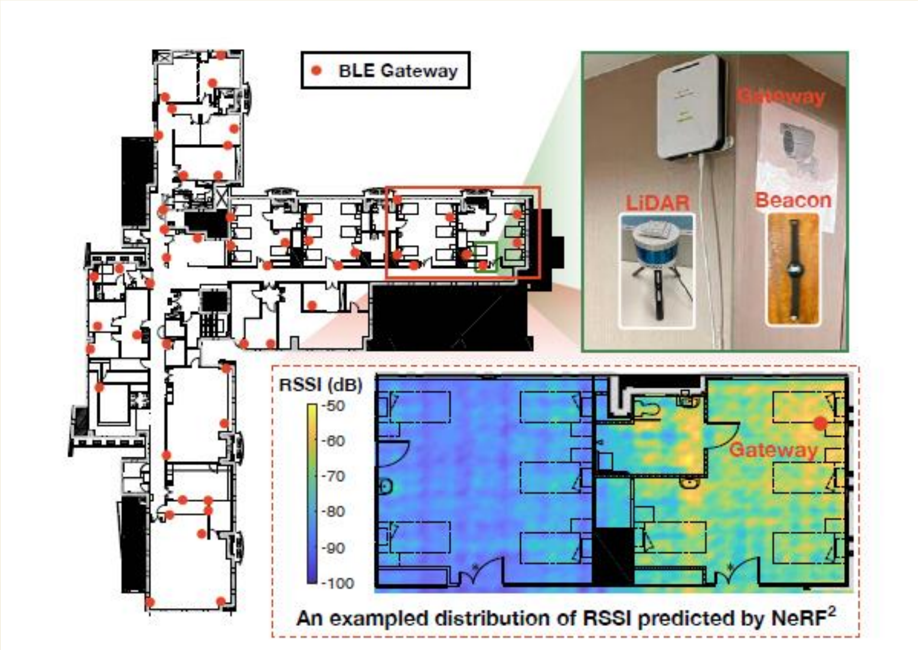
Mildenhall, Ben, et al. "Nerf: Representing scenes as neural radiance fields for view synthesis." *Communications of the ACM* 65.1 (2021): 99-106.

Analogy to RF channel:

With one Tx (object emitting light), I see how it looks like (channel) from some points of view (training Rx), what will it look like from other points not before seen from (testing Rx)?

Previous Work





Zhao, Xiaopeng, et al. "Nerf2: Neural radio-frequency radiance fields." *Proceedings of the 29th Annual International Conference on Mobile Computing and Networking*. 2023.

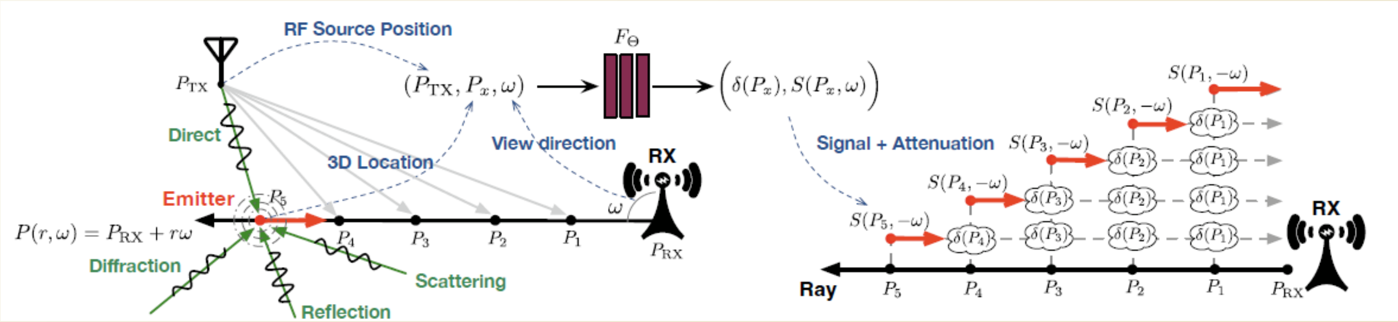
NeRF2:

1. Learns channel seen by a Rx, parameterized into function under spherical coordinates

2. One Tx, multiple RX locations, learn & do channel prediction

Previous Work

$$R(\omega) = \int_0^D \underbrace{\exp\left(\int_0^r \hat{\delta}(P(\tilde{r},\omega))\, d\tilde{r}\right)}_{\text{Attenuation Network}} \overbrace{S(P(r,\omega),-\omega)}^{\text{Radiance Network}}\ dr$$

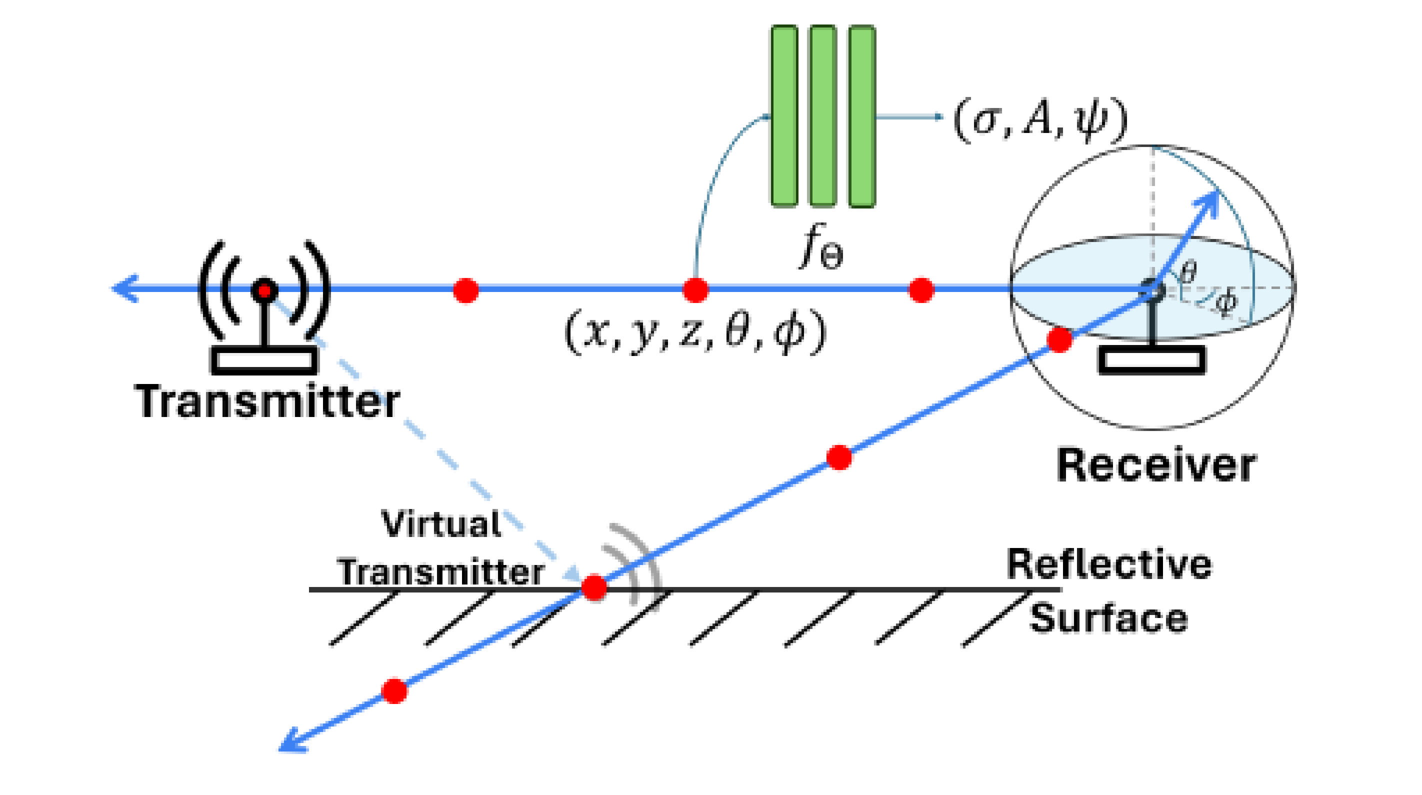


Zhao, Xiaopeng, et al. "Nerf2: Neural radio-frequency radiance fields." *Proceedings of the 29th Annual International Conference on Mobile Computing and Networking*. 2023.

NeRF2:

1. For each voxel, MLP learns the attenuation network and the radiance network, all the voxels along the ray contribute to the total channel seen by the Rx on the end of the ray

Previous Work



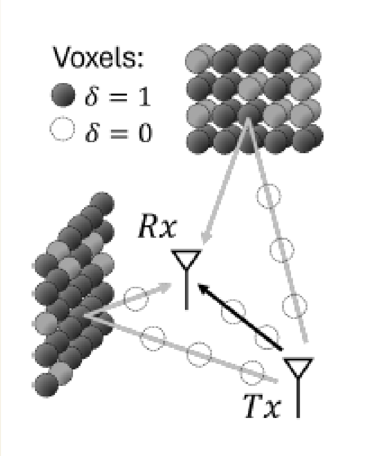
Lu, Haofan, et al. "Newrf: A deep learning framework for wireless radiation field reconstruction and channel prediction." *arXiv preprint arXiv:2403.03241* (2024).

NeWRF: Enhanced version of NeRF2, using full channel instead of power

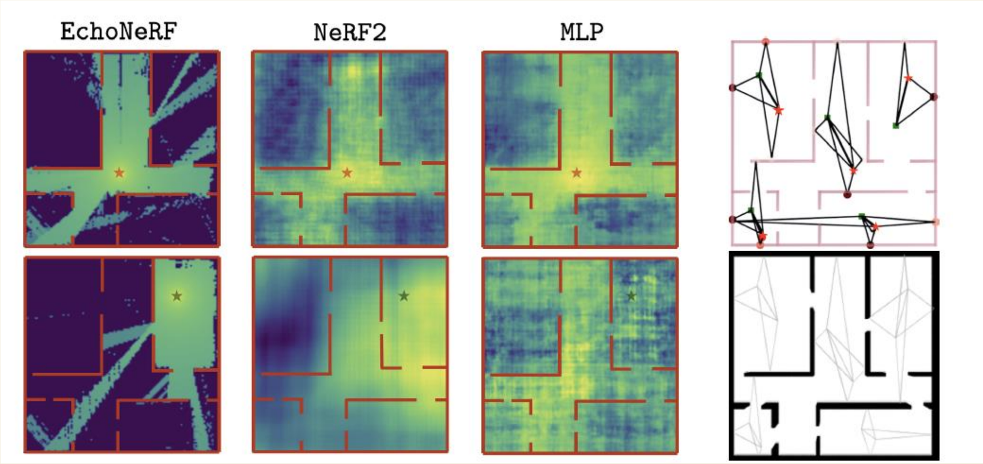
But neither NeRF2 or NeWRF solved one problem: what is the exact geometry of the environment (walls and objects), and how they affect the RF rays propagation paths, and after that, the accurate channel?

Previous Work

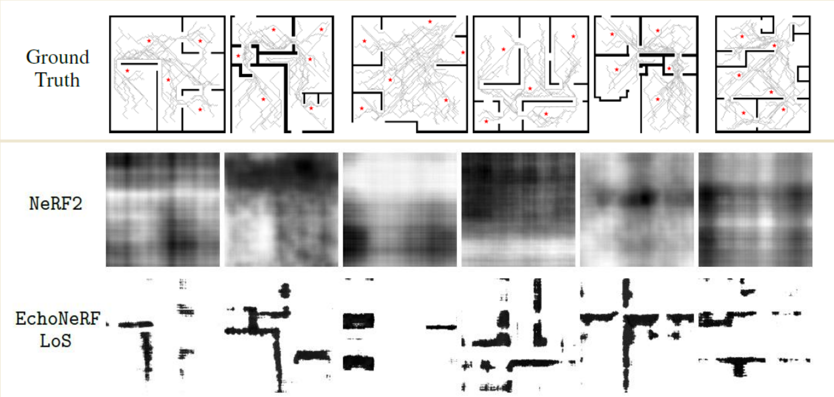
More precise model



RSSI (channel)heatmap



Room layout inference

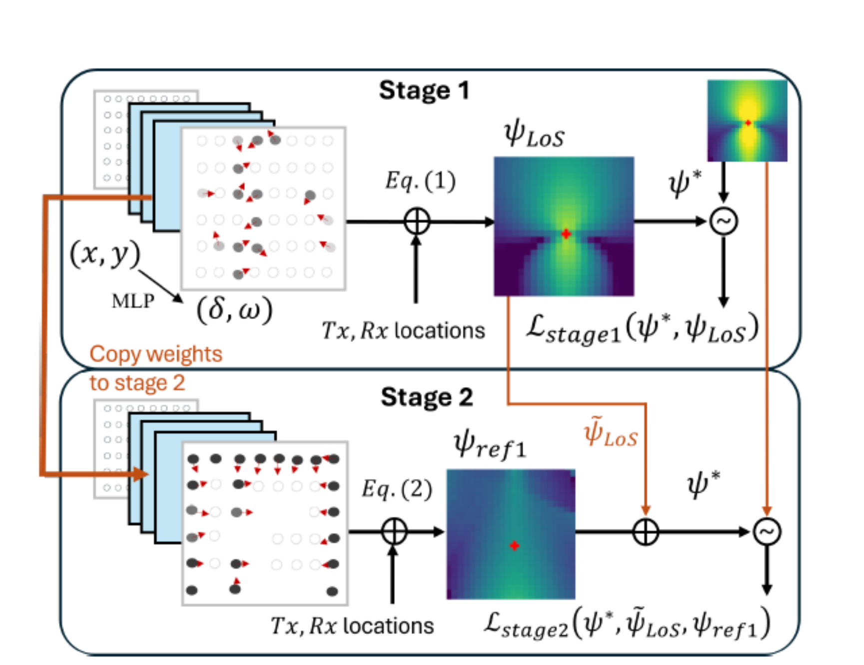


Amballa, Chaitanya, et al. "Can NeRFs See without Cameras?." *arXiv preprint arXiv:2505.22441* (2025).

EchoNeRF

1. Infers 2D scene model (room layout)
2. Does channel prediction

Previous Work



$$\psi_{LoS} = K \frac{\prod_{\{i|v_i \in LoS\}} (1-\delta_i)}{d^2}$$

$$\psi_{ref}(v_j) = \delta_j f(\theta, \beta) \frac{\prod_{k \in \{Rx:v_j\}} (1-\delta_k) \prod_{l \in \{v_j:Tx\}} (1-\delta_l)}{\left(d_{Tx:v_j} + d_{v_j:Rx}\right)^2}$$

Amballa, Chaitanya, et al. "Can NeRFs See without Cameras?." *arXiv preprint arXiv:2505.22441* (2025).

EchoNeRF

Two-stage MLP for LOS+Reflection

EchoNeRF is good for it can infer the indoor geometry, and that it leverages precise geometrical modelling to improve channel accuracy, but one crucial shortage is that it ignores the diffraction part.

Our Work

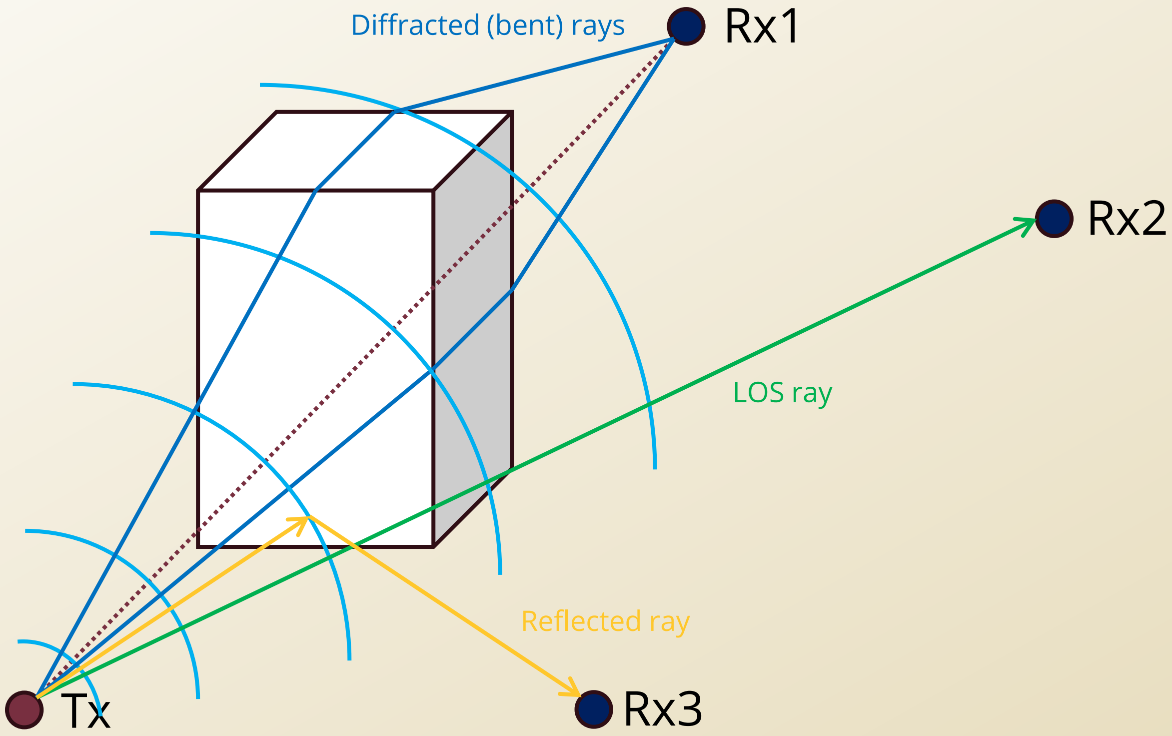
1. Infer 3D scene model
2. Do more accurate channel prediction

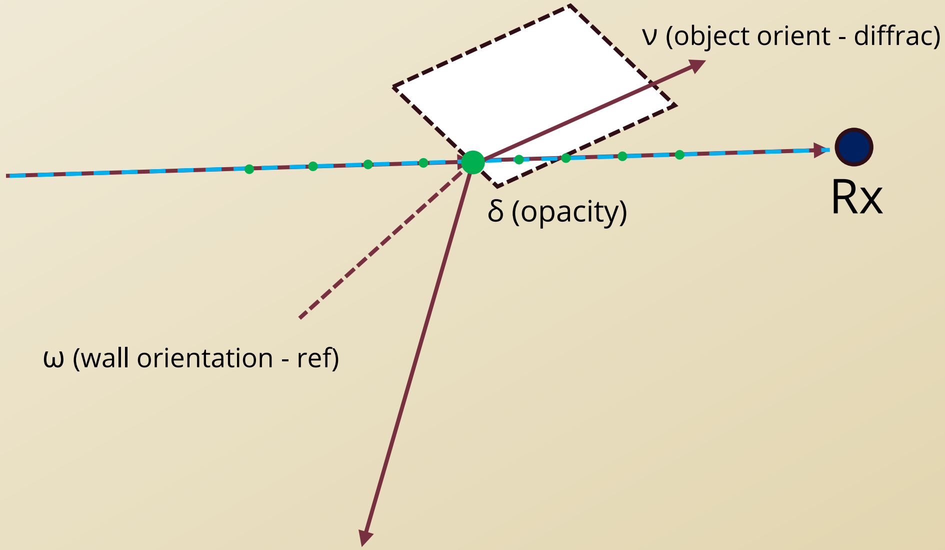
Cases:

1. With a portion of Rxs used for training with 1 Tx, predict channel received on the rest of the Rx locations

2. With all Rxs used for training with **more than 1** Tx, predict channel given a new Tx location

Methodology

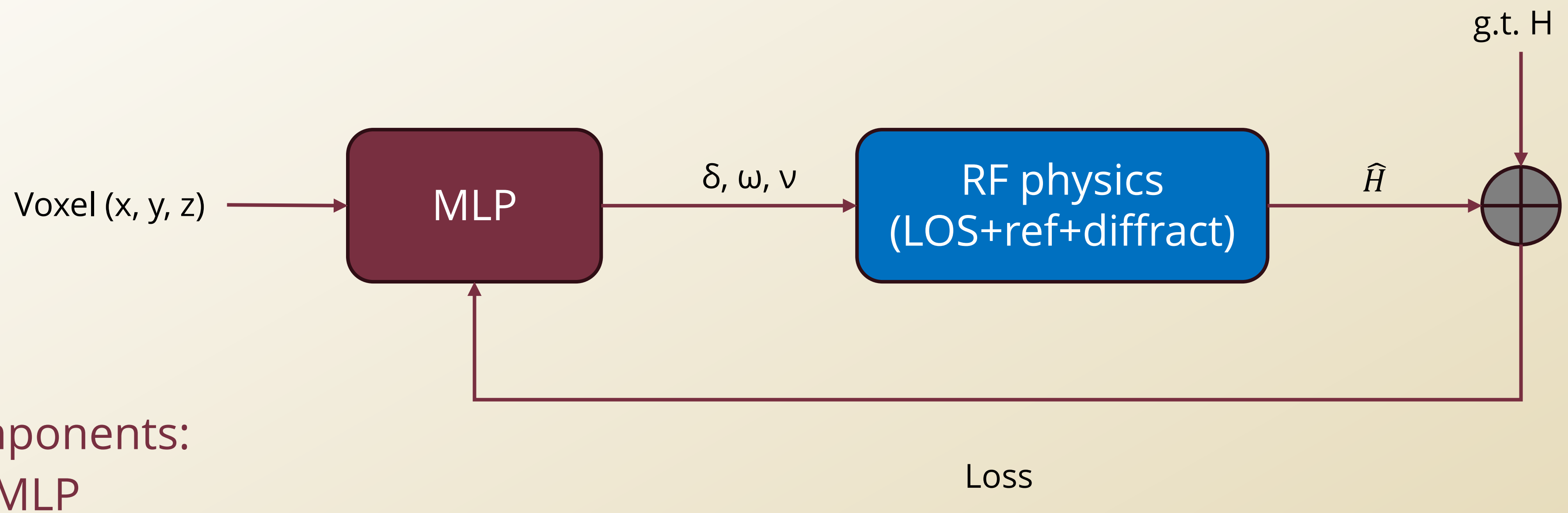




Voxel property to learn: δ (opacity), ω (wall orientation), ν (object orientation)

With all voxels learnt across the room, the channel seen by any Rx can be computed by physics model (simplified and inaccurate)

Methodology



MLP components:

- VoxelMLP
- EdgeMLP
- SurfaceMLP

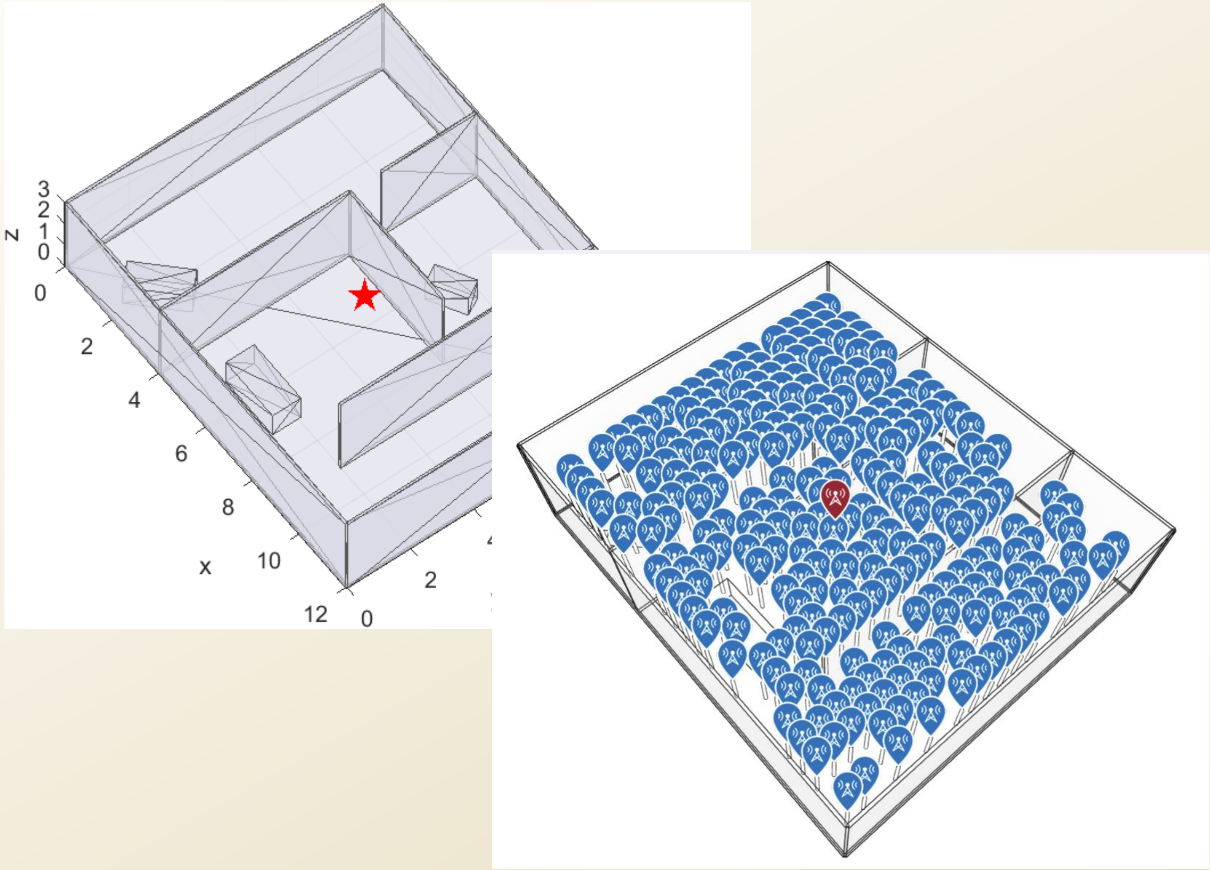
LOS uses VoxelMLP;

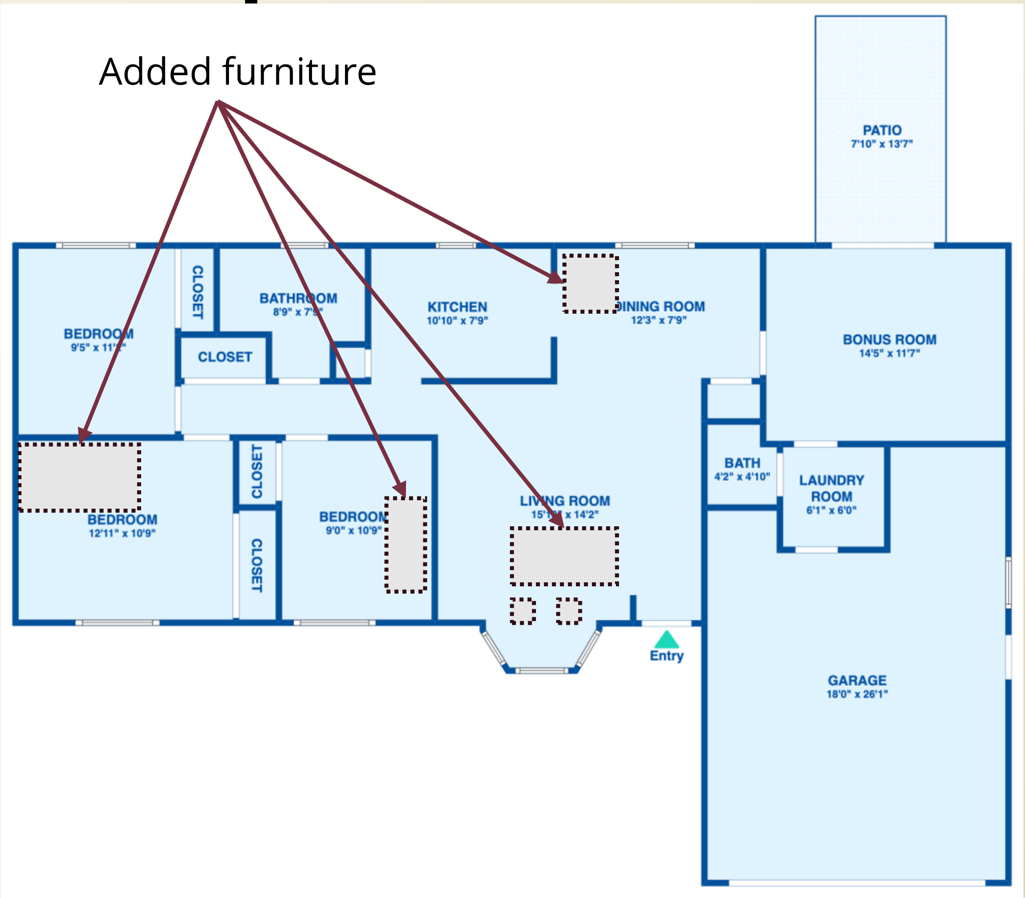
Reflection uses VoxelMLP;

Diffraction uses VoxelMLP+EdgeMLP+SurfaceMLP

Use this because LoS/reflection/diffraction all depend on the same environment occupancy/attenuation field. However, they are somehow coupled and can cause overfitting: e.g., where diffraction should not occur, the MLP still "make up" diffraction and degrade the accuracy, hence, partial knowledge of the scene might be needed (or not)

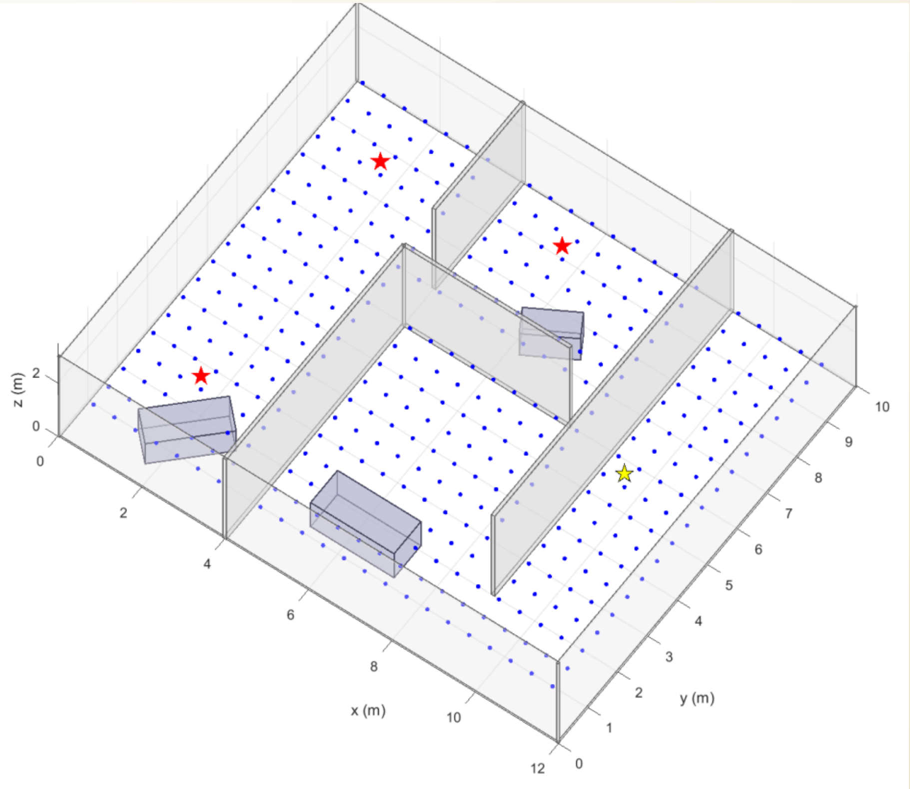
Dataset-case 1 (single Tx, partial Rxs)





1. Matlab simulated data
2. Modified real-world indoor dataset (Zillow)

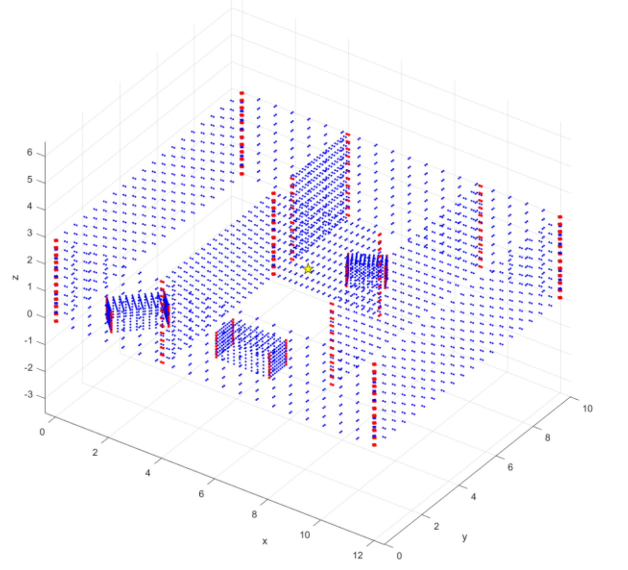
Dataset-case 2 (multi-Tx, all Rx)



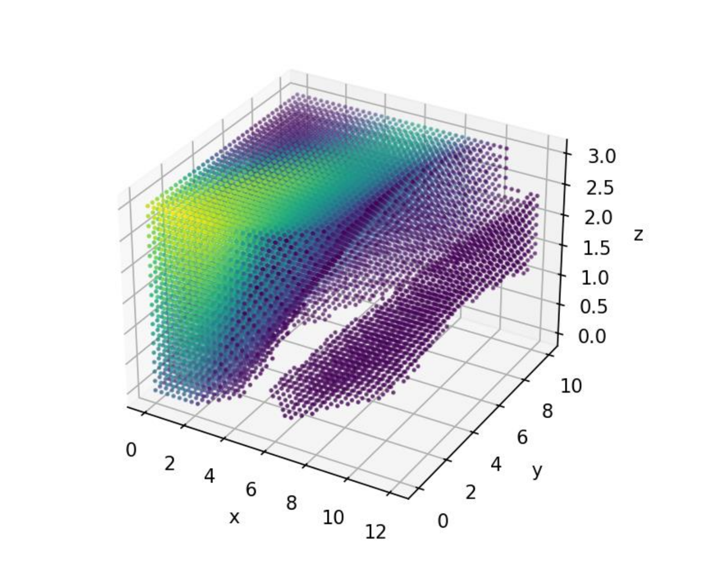
1. Matlab simulated data
2. Modified real-world indoor dataset

Experiments

1. Use Matlab to construct 3D indoor environment, and use Raytracing to get ground truth, then use MLP+physics model to train and do prediction,
2. Randomly place furniture in the Zillow floorplans, and make them all 3D, and use Sionna to get ground truth, then learn and predict



*Currently using assumed prior knowledge of scene to confine the diffraction part (the red edges), in real life might be gotten from SLAM, or even sketching by people who had a glance



*Current attempt to infer the indoor geometry without prior knowledge

Preliminary Results

```
Epoch 294 | loss = 1.677325
Epoch 295 | loss = 1.950391
Epoch 296 | loss = 1.674461
Epoch 297 | loss = 1.508498
Epoch 298 | loss = 1.562376
Epoch 299 | loss = 1.499096
Epoch 300 | loss = 1.609486
```

```
Epoch 294 | loss = 8.403820
Epoch 295 | loss = 8.207875
Epoch 296 | loss = 7.649276
Epoch 297 | loss = 7.708986
Epoch 298 | loss = 7.264952
Epoch 299 | loss = 8.354710
Epoch 300 | loss = 8.043865
```

- In training: With same training epoch, DiffracNeRF converges faster than EchoNeRF
- In training : Limit of performance:

  Validation Loss Plateaus at around

  Epoch 4000 | loss = 0.309098 (DiffracNeRF)

  Epoch 4000 | loss = 1.006198 (EchoNeRF)
- Ablation (take off surface part):

  Epoch 4000 | loss = 0.30 (DiffracNeRF no surface prop.)

Preliminary Results

Testing - case 1 (single Tx, partial Rxs):

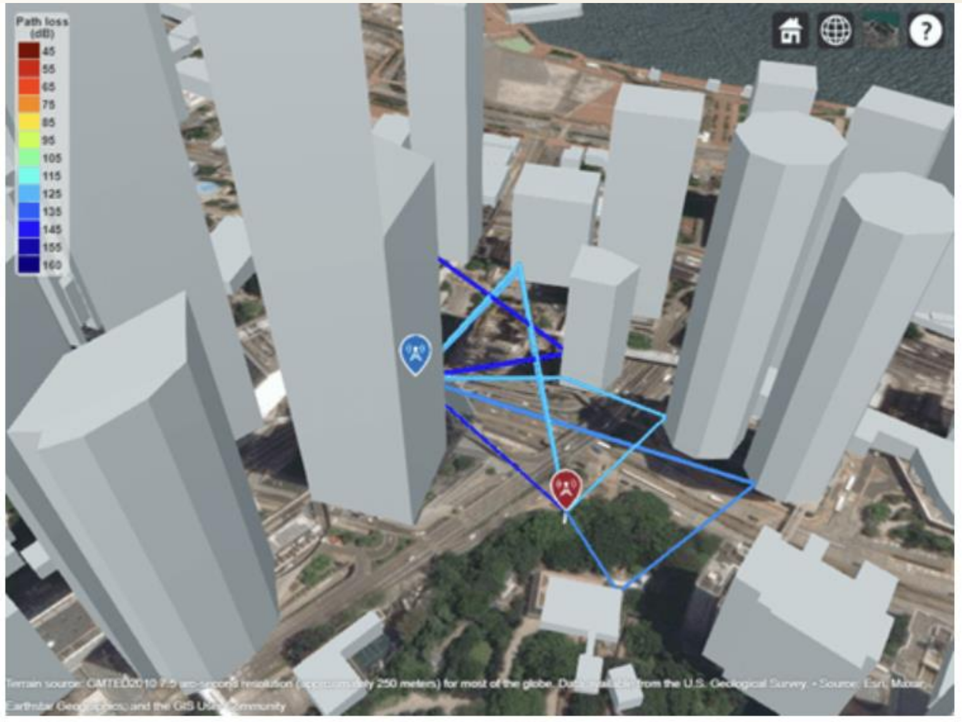
	EchoNeRF	DiffracNeRF	DiffracNeRF with partial scene info.
rmse_linear	1.990901e-06	2.255588e-06	2.160359e-06
mae_linear	6.806826e-07	7.455429e-07	6.996743e-07

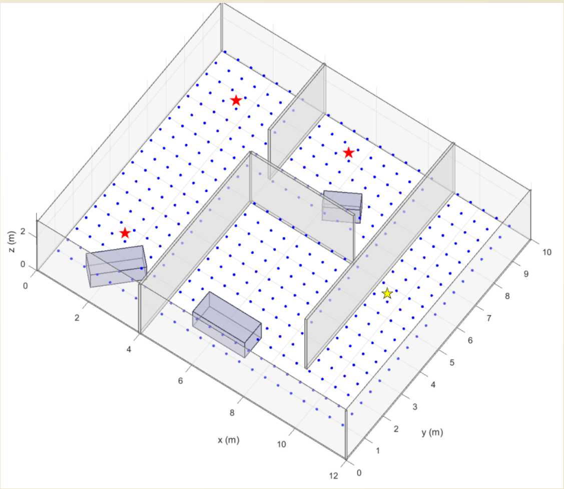
Preliminary Results

Testing - case 2 (multi-Tx, all Rx) :

	EchoNeRF	DiffracNeRF
rmse_linear	8.351536e-09	8.350917e-09
mae_linear	3.014521e-09	3.006533e-09

Prospective applications





- 5G/6G network: Channel prediction in clustered city environment
- 3D scene inference by RF only